



Paper 1 Writing Guide



Data at a Glance

330

Joint Measurements

Across 5 granodiorite road-cut outcrops,
Makkah Region

4/5

Sites Confirmed

Rayleigh $p < 0.05$ — non-uniformity
confirmed

This paper provides statistical proof of two distinct joint families co-existing at all five granodiorite road-cut outcrops in the Makkah Region, separating the Najd (NW–SE) and Red Sea (NE–SW) families by orientation and concentration.

Lithological uniformity across all sites (all granodiorite) isolates structural from lithological variation — a key methodological strength and the quantitative foundation for all subsequent papers in the series.

What This Paper Is — and Isn't

Understanding scope is essential before drafting begins.



Statistical Bimodality Proof

Statistical proof of two distinct joint families co-existing at all five sites — the paper's core deliverable



Family Separation

Separation of Najd (NW–SE) and Red Sea (NE–SW) families by orientation and concentration



Per-Site Parameters

Site-by-site mean direction, $\bar{R}$, and κ per family across all five outcrops



Geological Interpretation

Site 2 overprinting and Site 5 three-way mixture — the two key anomalies requiring dedicated discussion



Out of Scope

Najd Intensity Index (Paper 2), slope stability (Paper 3), and risk mapping (Papers 4–5) are explicitly excluded



This paper establishes the quantitative foundation that all subsequent papers depend on.

Two Tectonic Episodes

1

Najd Fault System (~600 Ma)

Proterozoic: Sinistral transpression along a NW–SE shear corridor imprints the persistent NW-trending joint set — the dominant family observed at all five sites.

2

Red Sea Rifting (~30 Ma)

Cenozoic: NE–SW extension creates the subordinate but locally dominant NE-trending set. Particularly strong at Site 2, where it overprints the Najd signal.

3

~570 Ma Separation

Implication: Two episodes so far apart in time produce a bimodal fabric that is geologically expected — not coincidental. This is the interpretive backbone of the paper.

Statistical Methods

1 Rayleigh Test

Non-uniformity: Tests non-uniformity of azimuths; $Z = n \cdot \bar{R}^2$; $p < 0.05$ confirms a preferred direction. Confirmed at 4 of 5 sites.

2 V-test

Targeted bearing: One-tailed variant targeting a specific bearing ($\sim 135^\circ$ Najd, $\sim 050^\circ$ Red Sea); useful where Rayleigh is borderline.

3 Von Mises Fitting

Parameters: μ = mean direction, κ = concentration; **10 fits** total (5 sites $\times$ 2 families). Cite Fisher (1993) or Mardia & Jupp.

📌 $\bar{R}$ **Mean Resultant Length** is the core diagnostic. Najd yields $\bar{R} = 0.98\text{--}1.00$ across sites — the key quantitative finding of the paper.

Key Results by Site

Sites 1, 3 & 4 — Canonical Pattern

Najd dominant: $\bar{R} \approx 0.98\text{--}1.00$, $\mu \approx 135^\circ$. Red Sea family subordinate with Rayleigh $p < 0.05$. Classic bimodal pattern confirmed.

Site 2 — Cenozoic Overprint

Red Sea dominant: Najd present but weaker; Red Sea $\bar{R}$ exceeds Najd with $p < 0.05$. Reflects Cenozoic reactivation near the rift.

Site 5 — Structural Transition Zone

Three-way mixture: 39% Najd, 18% Red Sea, 42% unclassified. Pooled Rayleigh $p = 0.228$. Family decomposition required before NII input.



Discussion: Three Interpretive Threads

Each thread connects the statistical output to a broader geological narrative.



Thread 1: Bimodal Expectation

- Two independent tectonic episodes should leave two joint sets
- Data confirms this at all five sites — as geologically predicted
- Najd NW–SE dominant; Red Sea NE–SW subordinate



Thread 2: Site 2 Overprint

- Stronger Red Sea coherence at Site 2
- Reflects Cenozoic reactivation or proximity to a rift-related structure
- Bimodal fabric persists but family dominance inverts



Thread 3: Site 5 Complexity

- Three-way mixture interpreted as a shear-zone boundary
- 42% of joints remain unclassified — pooled Rayleigh $p = 0.228$
- Family decomposition required before NII input in Paper 2

Figures Checklist

01

Figure 1 — Location Map

Five outcrops with Najd corridor and Red Sea rift axis inset

02

Figure 2 (2a–2e) — Rose Diagrams

Per site in 10° bins, colour-coded by family. Site 5 three-class decomposition — **the paper's core visual**

03

Table 2 — Summary Statistics

n , $\bar{R}$, μ , κ , and Rayleigh p for each family at each site — must be internally consistent with rose diagrams



Critical Gate

Fig 2 + Table 2

Must be finalised before drafting the Results section — internal consistency is non-negotiable



Figure 3 (Optional)

Stereonet of all 330 poles coloured by family — include if time allows

Paper Structure & Word Targets

Five sections with defined scope and word counts to guide drafting.

- **1 — Abstract (~150 w)**
Problem → data → methods → finding → significance for Paper 2
- **2 — Introduction (~400 w)**
Gap statement + two objectives
- **3 — Setting / Sites / Field Methods (~350 w + table + 250 w)**
Geological context, outcrop descriptions, measurement protocol
- **4 — Statistical Methods (~400 w)**
Cite Fisher (1993) or Mardia & Jupp. Rayleigh, V-test, and Von Mises derivations

 **Section 5 — Results → Discussion → Conclusions:** Site-by-site stats → three threads → four-point summary



CONCLUSIONS

Four Points to Land



Non-Uniformity Confirmed

Statistical result: Non-uniform distribution confirmed at 4 of 5 sites ($p < 0.05$), establishing directional preference for both families.



Site-Specific Anomalies

Key deviations: Site 2 Red Sea overprint and Site 5 structural complexity requiring family decomposition are identified and interpreted.



Bimodal Fabric Universal

Regional pattern: Bimodal fabric present at all sites — Najd NW–SE dominant, Red Sea NE–SW subordinate. Two tectonic episodes confirmed.

Target Journal

Journal of Structural Geology — submission in **3–5 months** from now. Data collection complete; ready to draft.